Validity Analysis: HOPE

Target context: US and South Asian Mental Health Counseling Practitioners (OMHC/Teletherapy)

Overall risk: **HIGH**

Deployment Context

Use case: A conversational AI system for mental health counseling classifies dialogue acts in real-time counseling conversations using the HOPE benchmark and any classification model. The system identifies the communicative intent (e.g., question, clarification) of each utterance by counselors and clients, and subsequently allows downstream tasks such as session quality assessment, automated counselor feedback, and training support in clinical and teletherapy settings.

Target population: Mental health counselors, trainee therapists, and clinical supervisors who conduct or review online counseling sessions in English. This includes practitioners and researchers working with teletherapy platforms and peer support services who need automated tools for counseling quality evaluation and dialogue understanding.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ⚠	2	Significant gaps	high
Input Content ⚠	2	Significant gaps	high
Input Form	3	Moderate gaps	medium
Output Ontology ✓	3	Moderate gaps	high
Output Content	3	Moderate gaps	medium
Output Form ✓	4	Minor gaps	high
Average	**2.8**		

⚠ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

HOPE provides a well-engineered dialogue-act classification benchmark for structured US clinical counseling (CBT, child, family therapy) with a thoughtfully designed 12-label taxonomy [Q17, Q27], substantial inter-annotator agreement [Q49], and rigorous evaluation methodology [Q63, Q87, Q88]. For the in-scope structured clinical therapy portion of the US deployment, the benchmark is reasonably valid. However, the deployment explicitly extends to peer support, crisis intervention, and South

Asian English contexts, and HOPE's source data, taxonomy, and annotators are predominantly aligned with US-based clinical therapy [Q98, Q99]. The HIGH-priority IO dimension shows a clear ontological gap: peer support communicative patterns and crisis-specific acts (safety-planning, risk stratification) have no dedicated representation and would default to overloaded categories like ACK or the residual GC catch-all [Q44, Q77]. No drop-in supplementary benchmark exists for crisis dialogue-act classification [WEB-2, WEB-20], and ESConv [WEB-18] / EMH [WEB-19] provide only structurally-different peer-support frameworks. Output form is well aligned (score 4); input form has bounded concerns from live-ASR fidelity (score 3); output ontology shows real boundary ambiguity even within scope (score 3); annotator representation lacks documented South Asian or peer/crisis presence (OC score 3). Input ontology and input content carry the largest validity risks (both score 2).

ID	Evidence
Q17	"We propose twelve dialogue-act labels that are aligned with mental-health counseling session." (p.2)
Q27	"Since the counseling conversations have inherent differences with the standard conversations (such as SwitchBoard dataset conversation), it demands a carefully designed set of dialogue-act labels capable of catering the requirements of counselling conversations. Hence, we, in consultation with therapist and counselling experts, design a set of 12 dialog-act labels that are arranged in a hierarchy." (p.3)
Q44	"General Chit-Chat (GC): Other utterances that do not belong to any of the above labels are tagged as GC, possibly because of the vagueness and the lack of sense in the context of the conversation." (p.4)
Q49	"We obtain the inter-rater agreement score of 0.7234 – which falls under the 'substantial' [7] category." (p.4)
Q63	"To measure the performances of SPARTA and other baseline systems, we compute macro-F1, weighted-F1, and accuracy scores." (p.6)
Q77	"We can observe that SPARTA consistently yields good scores for the majority of the dialogue-acts, except for the Acknowledgement (ACK) where it records F1-score of merely 47.86%." (p.7)
Q87	"Moreover, we also report the mean of the 3-fold cross-validation results for both SPARTA-MHA and SPARTA-TAA, and the results are consistent with the train-val-test split case." (p.7)
Q88	"We also perform statistical significance T-test comparing SPARTA-TAA and the best performing baseline (CASA). We observe that our results are significant with > 95% confidence across macro-F1 (p-value= 0.009), weighted-F1 (p-value= 0.014), and accuracy (p-value= 0.048) values." (p.7)
Q98	"Another important aspect of the current work is that the majority of the sessions in HOPE belong to the mental health professionals and patients based in United States." (p.8)
Q99	"Hence, the effectiveness of SPARTA on data from other geographical or demographical regions may vary." (p.8)
WEB-2	988 Lifeline three-phase crisis model uses communicative acts not represented in HOPE labels (https://988lifeline.org/wp-content/uploads/2023/02/FINAL_988_Suicide_and_Crisis_Lifeline_Suicide_Safety_Policy_-3.pdf)
WEB-18	ESConv 8-strategy framework for peer support is structurally distinct from HOPE (https://aclanthology.org/2021.acl-long.269/)
WEB-19	https://github.com/behavioral-data/Empathy-Mental-Health
WEB-20	Crisis Text Line research uses discourse analysis rather than HOPE-compatible dialogue-act labels; no crisis-specific dialogue-act benchmark identified (https://www.crisistextline.org/data-philosophy/research-impact/)

Practical Guidance

What This Benchmark Measures

HOPE measures the ability to assign one of 12 counseling-aligned dialogue-act labels to utterances in dyadic clinical therapy sessions, evaluating conversational-flow classification rather than clinical judgment [Q14, Q102]. For the deployment, this provides reasonable evidence about model performance on US-based structured clinical therapy (CBT/child/family) — particularly for tracking conversational trajectory in supervision and feedback workflows. Output Form (score 4) and Output Ontology design (score 3) are the strongest dimensions for this in-scope use case.

Construct Depth

The benchmark probes dialogue-act classification with reasonable depth for structured clinical therapy: 12.9K utterances across 212 sessions [Q13], substantial inter-annotator agreement [Q49], multi-tier baseline comparison [Q79, Q80], and per-label diagnostic information [Q76]. However, depth degrades sharply outside this scope: there is no empirical evidence about how labels behave on peer support register, no evaluation on crisis-context utterances, no per-session-type or per-label

Kappa breakdown, and no out-of-distribution diagnostic framework. The discarded emotion-annotation pilot [Q126, Q129] indicates the taxonomy intentionally excludes affective dimensions that matter for quality assessment.

What Else You Need

For peer support deployment: ESConv [WEB-18] and EMH [WEB-19] provide complementary annotation frameworks (support strategies, empathy mechanisms) that should be evaluated alongside HOPE-trained models — neither uses HOPE-compatible labels, so cross-framework validation is required. For crisis intervention deployment: no equivalent benchmark exists [WEB-20], so crisis-specific dialogue-act validation is a net-new task — the 988 Lifeline structured model [WEB-2] should inform a custom label extension. For South Asian deployment: expert elicitation from South Asian clinical psychologists is needed to validate label-boundary behavior under different formality/directiveness norms (gap_id 3, deferred). For real-time teletherapy: empirical comparison of YouTube vs. live-ASR transcripts is needed to quantify input-form drift (gap_id 5, deferred). For supervisory feedback applications: per-label safety-critical performance reporting (especially for IRQ/YNQ behavior on crisis-relevant utterances) should supplement macro-F1.

ID	Evidence
Q13	“The HOPE dataset contains ~ 12.9K utterances across 212 mental-health counseling sessions.” (p.2)
Q14	“Each utterance in the dataset is tagged with one of the 12 counseling-aligned dialogue-act labels (c.f. Section 3).” (p.2)
Q49	“We obtain the inter-rater agreement score of 0.7234 – which falls under the 'substantial' [7] category.” (p.4)
Q76	“We also present the label-wise performance of SPARTA in Table 4.” (p.7)
Q79	“Based on the type of modelling, we categorize the baselines into three groups – utterance-driven (*Ut*), utterance+global context driven (*Ut* + GC), and utterance + global context + speaker-aware driven (*Ut* + GC + SA).” (p.7)
Q80	“Comparatively, SPARTA incorporates local context in addition to utterance, global context, and speaker dynamics (*Ut* + LC + GC + SA).” (p.7)
Q102	“It is important to note that we do not make any diagnostic claims.” (p.8)
Q126	“In earlier phases of experiments, we also considered the role of emotion in dialogue act classification.” (p.11)
Q129	“Due to such imbalance, this data was not used in the final version of our proposed architecture.” (p.11)
WEB-2	988 Lifeline three-phase crisis model uses communicative acts not represented in HOPE labels (https://988lifeline.org/wp-content/uploads/2023/02/FINAL_988_Suicide_and_Crisis_Lifeline_Suicide_Safety_Policy_-3.pdf)
WEB-18	ESConv 8-strategy framework for peer support is structurally distinct from HOPE (https://aclanthology.org/2021.acl-long.269/)
WEB-19	https://github.com/behavioral-data/Empathy-Mental-Health
WEB-20	Crisis Text Line research uses discourse analysis rather than HOPE-compatible dialogue-act labels; no crisis-specific dialogue-act benchmark identified (https://www.crisistextline.org/data-philosophy/research-impact/)

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *HOPE* | ***Context:*** *US and South Asian Mental Health Counseling Practitioners (OMHC/Teletherapy)*

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The HOPE input ontology is explicitly scoped to dyadic, structured clinical counseling sessions (CBT, child, family) [Q22, Q26], with the taxonomy reflecting therapist-driven speaker asymmetry [Q125]. The deployment explicitly extends to peer support and crisis intervention, which involve communicative patterns the taxonomy was not designed to capture. Peer support contexts lack the formal clinical hierarchy embedded in the speaker-initiative/responsive structure, and crisis intervention involves safety-planning, de-escalation, and risk-stratification acts with no dedicated representation — these would default to IRQ, YNQ, or the residual GC catch-all [Q44]. The 988 Lifeline's structured three-phase crisis model (Explore/Clarify/Plan) does not map onto HOPE's flat 12-label taxonomy [WEB-2]. No crisis-specific dialogue-act benchmark exists to fill the gap [WEB-20]. ESConv's 8-strategy framework for peer/social support [WEB-18] is structurally different from HOPE's taxonomy, indicating that the field treats peer support as ontologically distinct. Given peer support and crisis are HIGH-priority deployment contexts, this is a substantial ontological gap. The user's elicitation acknowledges this directly (A1).

ID	Evidence
Q22	"In this section, we present our dialogue-act classification dataset, called HOPE. In total, we annotated ~ 12.9*K* utterances with 12 dialogue-act labels carefully designed to cater to the requirements of a counseling session." (p.3)
Q26	"The data collection process provides us 12.9*K* utterances from 212 counseling therapy sessions – all of them are dyadic conversations only." (p.3)
Q44	"General Chit-Chat (GC): Other utterances that do not belong to any of the above labels are tagged as GC, possibly because of the vagueness and the lack of sense in the context of the conversation." (p.4)
Q125	"This is particularly apparent as a majority portion of the counseling conversation is driven by the therapist, thus the speaker-initiative dialogue-acts have higher relevance with the therapist utterances." (p.11)
WEB-2	988 Lifeline three-phase crisis model uses communicative acts not represented in HOPE labels (https://988lifeline.org/wp-content/uploads/2023/02/FINAL_988_Suicide_and_Crisis_Lifeline_Suicide_Safety_Policy_-3.pdf)
WEB-18	ESConv 8-strategy framework for peer support is structurally distinct from HOPE (https://aclanthology.org/2021.acl-long.269/)
WEB-20	Crisis Text Line research uses discourse analysis rather than HOPE-compatible dialogue-act labels; no crisis-specific dialogue-act benchmark identified (https://www.crisistextline.org/data-philosophy/research-impact/)

Strengths

- Taxonomy is explicitly designed for counseling rather than borrowed from generic dialogue-act schemes, with hierarchical organization (initiative/responsive/general) tailored to therapy [Q27, Q29]
- Speaker-aware modeling (SPARTA) reflects the therapist-patient asymmetry that does hold within structured clinical therapy use cases in scope [Q125]
- The taxonomy is appropriate for the structured-therapy portion of the deployment (CBT/child/family in US contexts), which forms a substantial part of the practitioner workflow

ID	Evidence
Q27	"Since the counseling conversations have inherent differences with the standard conversations (such as SwitchBoard dataset conversation), it demands a carefully designed set of dialogue-act labels capable of catering the requirements of counselling conversations. Hence, we, in consultation with therapist and counselling experts, design a set of 12 dialog-act labels that are arranged in a hierarchy." (p.3)

Q29	“Each utterance in the dialogue belongs to one of the three categories – speaker initiative, speaker responsive, and general.” (p.3)
Q125	“This is particularly apparent as a majority portion of the counseling conversation is driven by the therapist, thus the speaker-initiative dialogue-acts have higher relevance with the therapist utterances.” (p.11)

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories include structured clinical therapy acts (covered), peer support acts (mutual disclosure, emotional co-regulation, informal acknowledgment), and crisis-specific acts (safety-planning probes, lethality assessment, de-escalation, risk stratification). The deployment scope explicitly includes all three [elicitation A1].

IO-2: Yes — peer support and crisis-specific dialogue-act categories are absent. HOPE's taxonomy is built for 'patient and therapist' dyads in 'social counselling' [Q7, Q8] and for 'mental-health counseling session' [Q17], with no peer or crisis coverage. The 988 Lifeline's distinct conversational phases [WEB-2] and ESConv's separate strategy taxonomy [WEB-18] confirm that peer/crisis acts require different representational frameworks.

IO-3: No clearly irrelevant categories were identified — all 12 labels remain at least nominally applicable to general counseling dialogue. The risk is underrepresentation, not irrelevance.

IO-4: Major content-validity gaps: (a) crisis-specific safety-planning and risk-assessment acts have no dedicated label and would be subsumed into IRQ/YNQ/GC, indistinguishably from routine acts; (b) peer support emotional co-regulation likely collapses into the already-overloaded ACK or residual GC [Q44]; (c) the speaker asymmetry baked into the taxonomy [Q125] does not map onto peer dyads.

ID	Evidence
Q7	“Unlike general goal-oriented dialogues, a conversation between a patient and a therapist is considerably implicit, though the objective of the conversation is quite apparent.” (p.1)
Q8	“The nature of conversations in a social counselling setting is particularly distinct as compared to a conventional chit-chat or goal-oriented conversations.” (p.1)
Q17	“We propose twelve dialogue-act labels that are aligned with mental-health counseling session.” (p.2)
Q22	“In this section, we present our dialogue-act classification dataset, called HOPE. In total, we annotated ~ 12.9*K* utterances with 12 dialogue-act labels carefully designed to cater to the requirements of a counseling session.” (p.3)
Q26	“The data collection process provides us 12.9*K* utterances from 212 counseling therapy sessions – all of them are dyadic conversations only.” (p.3)
Q44	“General Chit-Chat (GC): Other utterances that do not belong to any of the above labels are tagged as GC, possibly because of the vagueness and the lack of sense in the context of the conversation.” (p.4)
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WEB-20	Crisis Text Line research uses discourse analysis rather than HOPE-compatible dialogue-act labels; no crisis-specific dialogue-act benchmark identified (https://www.crisistextline.org/data-philosophy/research-impact/)

Information Gaps

- No empirical validation of how HOPE labels behave on peer support or crisis-register text
- No ontology defined for crisis-specific acts such as safety-planning probes or lethality assessment

Requires Expert Verification

- Whether peer support practitioners would recognize their communicative patterns in HOPE's taxonomy
- Whether crisis intervention staff would consider the IRQ/YNQ collapse acceptable for their workflow

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Remediation

Gap 1: No dedicated label coverage for crisis-specific (safety-planning, risk-stratification, de-escalation) or peer-support (mutual disclosure, emotional co-regulation) communicative acts; these collapse into IRQ/YNQ/GC.

Recommendation: Before deploying in crisis or peer-support contexts, extend the taxonomy with crisis-specific labels informed by the 988 Lifeline three-phase model [WEB-2] and peer-support strategy labels informed by ESConv [WEB-18]. Validate the extended taxonomy on a held-out sample of crisis and peer-support transcripts before integrating into production feedback workflows.

ID	Evidence
WEB-2	988 Lifeline three-phase crisis model uses communicative acts not represented in HOPE labels (https://988lifeline.org/wp-content/uploads/2023/02/FINAL_988_Suicide_and_Crisis_Lifeline_Suicide_Safety_Policy_-3.pdf)
WEB-18	ESConv 8-strategy framework for peer support is structurally distinct from HOPE (https://aclanthology.org/2021.acl-long.269/)

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *HOPE* | ***Context:*** *US and South Asian Mental Health Counseling Practitioners (OMHC/Teletherapy)*

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

HOPE's source corpus is YouTube-sourced US-based counseling sessions [Q23, Q98], with the authors explicitly warning that 'the effectiveness of SPARTA on data from other geographical or demographical regions may vary' [Q99]. The deployment extends to South Asian (India and broader) contexts, where formality and counselor-directiveness norms may differ — though the user judges core label applicability to be largely intact (elicitation A2). YouTube selection bias likely suppresses disfluencies and false starts characteristic of real-time teletherapy [Q93]. Sessions are notably long (~59 utterances; therapist ~124 words/utterance) [Q56, Q57], reflecting structured formal therapy rather than peer support's shorter-turn register. No South Asian sessions or peer support sessions are present in the corpus. The South Asian deployment is HIGH-priority by user assessment but ranked MODERATE in the elicitation weights, since the user views label-level transfer as plausible. Combined gaps in geographic representation and session-type representation justify a score of 2.

ID	Evidence
Q23	“One of major hurdles we faced in the process of data collection was the unavailability of public counseling sessions, mainly due to the fact that they usually contain sensitive personal information. To curate this data, we carefully explored the web and collected publicly-available pre-recorded counselling videos on YouTube.” (p.3)
Q56	“In contrast to the regular patient-doctor conversations (e.g., SOAP), the dialogue sessions in HOPE are usually lengthy (~ 59 utterances per session).” (p.5)
Q57	“Moreover, the utterances in these sessions are themselves significantly longer as compared to other conversational datasets, with the average length of utterance of a patient being 103 words, whereas for therapist, it is ~ 124 words.” (p.5)
Q93	“We aim to tackle a very sensitive and a pervasive public-health crisis. We transcribe the data from publicly available counselling videos.” (p.8)
Q98	“Another important aspect of the current work is that the majority of the sessions in HOPE belong to the mental health professionals and patients based in United States.” (p.8)
Q99	“Hence, the effectiveness of SPARTA on data from other geographical or demographical regions may vary.” (p.8)

Strengths

- Reasonable scale (~12.9K utterances across 212 sessions) and balanced patient/therapist distribution [Q53, Q54]
- Authors transparently acknowledge US-centricity and explicitly caveat geographical/demographic generalization [Q98, Q99]
- Confidentiality handled via synthetic name assignment [Q24] and systematic name masking [Q97]

ID	Evidence
Q24	“To ensure confidentiality, we randomly assign synthetic names to all patients and therapists in all examples.” (p.3)
Q53	“In total, HOPE has transcripts for 12.9*k* utterances which are annotated with 12 dialogue-act labels.” (p.5)
Q54	“These utterances are evenly distributed between the patients and therapists with 6.38*K* and 6.47*K* utterances, respectively.” (p.5)
Q97	“The names of the patients and therapists involved in these sessions have been systematically masked.” (p.8)
Q98	“Another important aspect of the current work is that the majority of the sessions in HOPE belong to the mental health professionals and patients based in United States.” (p.8)
Q99	“Hence, the effectiveness of SPARTA on data from other geographical or demographical regions may vary.” (p.8)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — South Asian English clinical communication, including formality norms and counselor directiveness, may differ from US YouTube-sourced content. The user identifies this as a real but not acute concern (elicitation A2). National Mental Health Survey of India documented 75–93% treatment gaps and significant urban-rural heterogeneity [WEB-1], suggesting India-context delivery patterns differ materially from US.

IC-2: Partial mismatch — content reflects US clinical communication norms and structured therapy register. Peer support and crisis register are entirely absent; South Asian register is unrepresented.

IC-3: Yes — sessions encode US counselor directiveness and therapeutic framing conventions [Q98]. The extent to which these transfer to South Asian English clinical settings is documented as a deferred verification item in the regional context.

IC-4: INSUFFICIENT DOCUMENTATION — no regional annotators from South Asia or peer/crisis settings were recruited; geographic origin of annotators not specified beyond institutional affiliation.

IC-5: Documented issues: (a) US-centricity acknowledged by authors [Q98, Q99]; (b) YouTube selection bias likely suppresses ASR-relevant phenomena like disfluencies/false starts [Q93]; (c) absence of peer support and crisis intervention content; (d) sessions are atypically long [Q56, Q57], representing formal therapy rather than the shorter-turn dynamics of OMHC peer/crisis exchanges.

ID	Evidence
Q56	"In contrast to the regular patient-doctor conversations (e.g., SOAP), the dialogue sessions in HOPE are usually lengthy (~ 59 utterances per session)." (p.5)
Q57	"Moreover, the utterances in these sessions are themselves significantly longer as compared to other conversational datasets, with the average length of utterance of a patient being 103 words, whereas for therapist, it is ~ 124 words." (p.5)
Q93	"We aim to tackle a very sensitive and a pervasive public-health crisis. We transcribe the data from publicly available counselling videos." (p.8)
Q98	"Another important aspect of the current work is that the majority of the sessions in HOPE belong to the mental health professionals and patients based in United States." (p.8)
Q99	"Hence, the effectiveness of SPARTA on data from other geographical or demographical regions may vary." (p.8)
WEB-1	Indian rural connectivity and resource gaps documented in National Mental Health Survey context (https://pmc.ncbi.nlm.nih.gov/articles/PMC7736743/)

Information Gaps

- Empirical comparison of South Asian English vs. US English counseling discourse [deferred — likely unsearchable]
- Quantitative comparison of YouTube-sourced vs. live teletherapy ASR transcripts for mental-health sessions
- Demographic breakdown of HOPE session participants beyond US-majority claim

Requires Expert Verification

- Whether South Asian clinical psychologists view core HOPE label boundaries as preserved under Indian English directiveness norms

- Whether peer support and crisis content patterns differ enough from clinical content to invalidate model behavior trained on HOPE

Remediation

Gap 1: Source corpus is overwhelmingly US-based [Q98] with no South Asian sessions and no peer-support or crisis-intervention content.

Recommendation: Collect and annotate a supplementary evaluation set of South Asian English clinical sessions, OMHC peer-support transcripts, and crisis-intervention exchanges. Use this set strictly for validation (not training) to estimate performance drift relative to HOPE's reported metrics.

ID	Evidence
Q98	“Another important aspect of the current work is that the majority of the sessions in HOPE belong to the mental health professionals and patients based in United States.” (p.8)

Input Form (IF)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: *HOPE* | ***Context:*** *US and South Asian Mental Health Counseling Practitioners (OMHC/Teletherapy)*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Both benchmark and deployment are text-only English with Latin script — no script or modality mismatch. ASR pipeline (OTTER + manual correction) [Q25] is text-aligned with deployment's teletherapy ASR transcripts. However, the benchmark text is hand-corrected for spelling and grammatical errors [Q25] and the authors acknowledge information loss in the speech-to-text pipeline [Q94], whereas live teletherapy transcripts will likely retain greater disfluency density, false starts, and overlapping turns. This is a latent form mismatch that the benchmark documentation does not address. Infrastructure assumptions are aligned for US contexts but India rural connectivity may degrade ASR quality [WEB-16, WEB-1]. Given the user's LOWER priority for IF and the absence of script/modality mismatch, a moderate score of 3 reflects the meaningful but bounded gap.

ID	Evidence
Q25	"In the next step, we extract the transcriptions of each video using OTTER (https://www.otter.ai/), an automatic speech recognition tool. Subsequently, we correct transcription errors to remove any noise (i.e., spelling or grammatical mistakes)." (p.3)
Q94	"The automatic transfer of utterance from the speech modality to the text causes some information loss, though we tried our best to recover them through manual intervention." (p.8)
WEB-1	Indian rural connectivity and resource gaps documented in National Mental Health Survey context (https://pmc.ncbi.nlm.nih.gov/articles/PMC7736743/)
WEB-16	Tele MANAS implementation faces technology glitches and uneven rural access (https://cmhlp.org/imho/blog/indias-digital-mental-health-landscape-government-initiatives-and-challenges/)

Strengths

- No script or modality mismatch — both benchmark and deployment operate on English Latin-script text [Q25]
- Standard tokenization-friendly input via pretrained RoBERTa [Q61, Q111] is widely supported in deployment infrastructure
- Documented systematic name masking for privacy [Q97] aligns with HIPAA/DPDPA expectations

ID	Evidence
Q25	"In the next step, we extract the transcriptions of each video using OTTER (https://www.otter.ai/), an automatic speech recognition tool. Subsequently, we correct transcription errors to remove any noise (i.e., spelling or grammatical mistakes)." (p.3)
Q61	"For each utterance *Ut* in a conversation dialogue, we extract the semantic representation through pre-trained RoBERTa model [25], which is subsequently utilized to leverage the local (*Lt*) and global (*Gt*) contextual information within the dialogue." (p.5)
Q97	"The names of the patients and therapists involved in these sessions have been systematically masked." (p.8)
Q111	"We use pretrained RoBERTa available from the Hugginface14 transformers for speaker aware as well as speaker invariant utterance representations." (p.10)

Checklist

Checklist item definitions:

ID	Assessment Question

IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Partial mismatch — HOPE used hand-corrected OTTER transcripts [Q25], while live teletherapy ASR will not benefit from manual correction. Disfluency, false-start, and overlapping-turn distributions will likely differ.

IF-2: US infrastructure adequate; India rural connectivity remains uneven [WEB-1, WEB-16]. Tele MANAS implementation challenges include 'overworked personnel, technology glitches, and uneven rural access' [WEB-16].

IF-3: Real-time teletherapy ASR vs. recorded YouTube ASR is the key domain-specific difference. Authors acknowledge information loss generally [Q94] but do not quantify the gap.

IF-4: Documented form concerns: (a) hand-correction step in HOPE [Q25] not available in live deployment; (b) YouTube source data may be lightly edited or pre-selected [Q93]; (c) no quantitative comparison available between YouTube and real-time clinical ASR fidelity.

ID	Evidence
Q25	“In the next step, we extract the transcriptions of each video using OTTER (https://www.otter.ai/), an automatic speech recognition tool. Subsequently, we correct transcription errors to remove any noise (i.e., spelling or grammatical mistakes).” (p.3)
Q93	“We aim to tackle a very sensitive and a pervasive public-health crisis. We transcribe the data from publicly available counselling videos.” (p.8)
Q94	“The automatic transfer of utterance from the speech modality to the text causes some information loss, though we tried our best to recover them through manual intervention.” (p.8)
WEB-1	Indian rural connectivity and resource gaps documented in National Mental Health Survey context (https://pmc.ncbi.nlm.nih.gov/articles/PMC7736743/)
WEB-16	Tele MANAS implementation faces technology glitches and uneven rural access (https://cmhlp.org/imho/blog/indias-digital-mental-health-landscape-government-initiatives-and-challenges/)

Information Gaps

- Quantitative comparison of YouTube-sourced vs. real-time teletherapy ASR fidelity for mental-health sessions [deferred]
- Platform-specific ASR quality benchmarks for OMHC platforms (proprietary, not searchable)

Requires Expert Verification

- Whether disfluency-rich live ASR transcripts degrade SPARTA/HOPE-trained model performance materially

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Remediation

Gap 1: HOPE uses hand-corrected ASR transcripts [Q25]; live teletherapy ASR will retain disfluencies and errors that the benchmark does not represent.

Recommendation: Run an A/B comparison of model performance on hand-corrected vs. raw ASR samples from the target deployment platforms. If material degradation is observed, consider ASR post-processing, disfluency-aware fine-tuning, or confidence thresholding before surfacing labels to supervisors.

ID	Evidence
Q25	“In the next step, we extract the transcriptions of each video using OTTER (https://www.otter.ai/), an automatic speech recognition tool. Subsequently, we correct transcription errors to remove any noise (i.e., spelling or grammatical mistakes).” (p.3)

Output Ontology (OO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: *HOPE* | ***Context:*** *US and South Asian Mental Health Counseling Practitioners (OMHC/Teletherapy)*

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The 12-label taxonomy is explicitly 'aligned with mental-health counseling session' [Q17, Q104] and was designed in consultation with therapists [Q27]. The user judges the taxonomy fit-for-purpose for conversational flow tracking (elicitation A3). However, internal taxonomy stress is documented: ACK underperforms at F1=47.86% [Q77], indicating semantic overload even within clinical scope; confusion analysis shows YNQ:IRQ (26%), OD:ID (43%), ID:ACK (28%) error rates [Q91]; complementary label pairs are 'annotation-boundary-challenging' [Q51, Q52]. For the deployment's peer support and crisis extensions, the GC catch-all [Q44] becomes the de facto repository for any communicative act outside the clinical frame, and crisis-specific acts (safety-planning, risk stratification) lack dedicated labels — collapsing into IRQ/YNQ with potentially safety-critical consequences. The discarded emotion annotation pilot [Q126, Q127, Q129] indicates the authors recognized affective dimensions but chose not to incorporate them. Output is single-label per utterance [Q30], which is appropriate for the framing but limits expressiveness for utterances spanning multiple acts. Score of 3 reflects fit-for-purpose taxonomy for in-scope clinical use but real boundary ambiguity at deployment edges.

ID	Evidence
Q17	“We propose twelve dialogue-act labels that are aligned with mental-health counseling session.” (p.2)
Q27	“Since the counseling conversations have inherent differences with the standard conversations (such as SwitchBoard dataset conversation), it demands a carefully designed set of dialogue-act labels capable of catering the requirements of counselling conversations. Hence, we, in consultation with therapist and counselling experts, design a set of 12 dialog-act labels that are arranged in a hierarchy.” (p.3)
Q30	“Sometimes, an utterance can have multiple dialog-acts; however, they are rare in the annotated dataset. Hence, for simplicity, we consider only one (primary) label for each utterance.” (p.3)
Q44	“General Chit-Chat (GC): Other utterances that do not belong to any of the above labels are tagged as GC, possibly because of the vagueness and the lack of sense in the context of the conversation.” (p.4)
Q51	“The broad definitions of speaker-initiative and speaker-responsive dialogue-act label pairs, IRQ & ID; ORQ & OD, and CRQ & CD, seem complementary in nature.” (p.4)
Q52	“However, the dataset does not support the above view entirely.” (p.4)
Q77	“We can observe that SPARTA consistently yields good scores for the majority of the dialogue-acts, except for the Acknowledgement (ACK) where it records F1-score of merely 47.86%.” (p.7)
Q91	“We observe three pairs with significant error-rates (≥ 25%) – YNQ:IRQ (26%), OD:ID (43%), ID:ACK (28%).” (p.7)
Q104	“We defined twelve dialogue-act labels to cater to the requirement of counselling sessions.” (p.8)
Q126	“In earlier phases of experiments, we also considered the role of emotion in dialogue act classification.” (p.11)
Q127	“We annotated a subset of our dataset with three emotion classes consisting of 'positive', 'negative' and 'neutral'.” (p.11)
Q129	“Due to such imbalance, this data was not used in the final version of our proposed architecture.” (p.11)

Strengths

- Taxonomy designed in consultation with therapists and counseling experts [Q27], grounded in counseling-session requirements rather than borrowed
- Three-tier hierarchical structure (initiative/responsive/general) [Q29, Q32] provides interpretable organization

- Per-label F1 reporting [Q76] surfaces label-level performance limitations transparently
- Authors explicitly disclaim diagnostic intent [Q102], appropriate for a flow-classification scope

ID	Evidence
Q27	"Since the counseling conversations have inherent differences with the standard conversations (such as SwitchBoard dataset conversation), it demands a carefully designed set of dialogue-act labels capable of catering the requirements of counselling conversations. Hence, we, in consultation with therapist and counselling experts, design a set of 12 dialog-act labels that are arranged in a hierarchy." (p.3)
Q29	"Each utterance in the dialogue belongs to one of the three categories – speaker initiative, speaker responsive, and general." (p.3)
Q32	"Our annotation scheme assigns three distinct dialogue-act labels to the first two categories, while the remaining four labels belong to the general category." (p.4)
Q76	"We also present the label-wise performance of SPARTA in Table 4." (p.7)
Q102	"It is important to note that we do not make any diagnostic claims." (p.8)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Labels are nominally regionally relevant (greeting, information delivery, etc. are universal) but encode US clinical session structure assumptions [Q98]. South Asian directiveness norms may shift label-boundary behavior.

OO-2: Missing categories: (a) crisis-specific safety-planning and risk-assessment acts; (b) peer support emotional co-regulation acts; (c) empathic reflection vs. simple acknowledgment distinction (the user notes this is an intentional out-of-scope choice — A3); (d) affective dimension was piloted but discarded due to imbalance [Q126, Q127, Q129].

OO-3: GC as a residual catch-all [Q44] effectively encodes 'not part of the structured clinical frame' as a category, which biases against acts characteristic of peer/crisis registers. Speaker asymmetry [Q125] presupposes therapist-patient hierarchy not present in peer dyads.

OO-4: Stakeholder-driven taxonomy redesign is needed for crisis-context deployment specifically; for in-scope clinical and South Asian extensions, the user judges core labels acceptable (A3).

OO-5: Documented issues: ACK overload [Q77], complementary-pair confusion [Q51, Q52, Q91], lack of crisis or peer support labels, abandoned emotion dimension, and the GC residual category becoming a semantic black hole for out-of-frame utterances.

ID	Evidence
Q17	"We propose twelve dialogue-act labels that are aligned with mental-health counseling session." (p.2)
Q27	"Since the counseling conversations have inherent differences with the standard conversations (such as SwitchBoard dataset conversation), it demands a carefully designed set of dialogue-act labels capable of catering the requirements of counselling conversations. Hence, we, in consultation with therapist and counselling experts, design a set of 12 dialog-act labels that are arranged in a hierarchy." (p.3)
Q44	"General Chit-Chat (GC): Other utterances that do not belong to any of the above labels are tagged as GC, possibly because of the vagueness and the lack of sense in the context of the conversation." (p.4)
Q51	"The broad definitions of speaker-initiative and speaker-responsive dialogue-act label pairs, IRQ & ID; ORQ & OD, and CRQ & CD, seem complementary in nature." (p.4)

Q52	“However, the dataset does not support the above view entirely.” (p.4)
Q77	“We can observe that SPARTA consistently yields good scores for the majority of the dialogue-acts, except for the Acknowledgement (ACK) where it records F1-score of merely 47.86%.” (p.7)
Q91	“We observe three pairs with significant error-rates (≥ 25%) – YNQ:IRQ (26%), OD:ID (43%), ID:ACK (28%).” (p.7)
Q98	“Another important aspect of the current work is that the majority of the sessions in HOPE belong to the mental health professionals and patients based in United States.” (p.8)
Q102	“It is important to note that we do not make any diagnostic claims.” (p.8)
Q125	“This is particularly apparent as a majority portion of the counseling conversation is driven by the therapist, thus the speaker-initiative dialogue-acts have higher relevance with the therapist utterances.” (p.11)
Q126	“In earlier phases of experiments, we also considered the role of emotion in dialogue act classification.” (p.11)
Q127	“We annotated a subset of our dataset with three emotion classes consisting of 'positive', 'negative' and 'neutral'.” (p.11)
Q129	“Due to such imbalance, this data was not used in the final version of our proposed architecture.” (p.11)
WEB-2	988 Lifeline three-phase crisis model uses communicative acts not represented in HOPE labels (https://988lifeline.org/wp-content/uploads/2023/02/FINAL_988_Suicide_and_Crisis_Lifeline_Suicide_Safety_Policy_-3.pdf)
WEB-18	ESConv 8-strategy framework for peer support is structurally distinct from HOPE (https://aclanthology.org/2021.acl-long.269/)

Information Gaps

- Per-label inter-annotator agreement breakdown (only aggregate Kappa reported)
- How GC and ACK absorb peer/crisis acts in practice — no empirical validation

Requires Expert Verification

- Whether crisis intervention staff can operate with IRQ/YNQ collapse for safety-planning probes
- Whether peer support practitioners would adopt the speaker-asymmetric taxonomy

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Remediation

Gap 1: ACK label is semantically overloaded (F1=47.86%) [Q77] and GC functions as a residual catch-all [Q44]; per-label and per-session-type Kappa breakdowns are unavailable.

Recommendation: Request or compute per-label and per-session-type inter-annotator agreement on a re-annotated subset. For deployment, suppress or flag (rather than display) ACK and GC predictions in supervisory dashboards until reliability is independently validated.

ID	Evidence
Q44	“General Chit-Chat (GC): Other utterances that do not belong to any of the above labels are tagged as GC, possibly because of the vagueness and the lack of sense in the context of the conversation.” (p.4)
Q77	“We can observe that SPARTA consistently yields good scores for the majority of the dialogue-acts, except for the Acknowledgement (ACK) where it records F1-score of merely 47.86%.” (p.7)

Output Content (OC)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: *HOPE* | ***Context:*** *US and South Asian Mental Health Counseling Practitioners (OMHC/Teletherapy)*

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Annotation was performed by three linguists with calibration and an expert-therapist moderator [Q45, Q46, Q47], with mental health professional consultation during guideline preparation [Q95]. Aggregate Cohen's Kappa = 0.7234 (substantial) [Q49] is reasonable. The user judges the convergent-validity risk low because dialogue-act labels are communicative rather than clinical (elicitation A4). However, there are real concerns: no per-label or per-session-type Kappa breakdown is reported, so agreement on overloaded labels (ACK, GC) is unknown; no documented South Asian, peer support, or crisis worker participation in annotation; annotator demographics are limited to age and experience range [Q50] with no geographic or cultural background reported; authors acknowledge 'annotator's bias cannot be ruled out completely' [Q96]. Given the user's MODERATE priority on OC and the partial documentation, a score of 3 reflects substantial agreement on a US-clinical-centric annotation framework with unverified cross-cultural and cross-session-type calibration.

ID	Evidence
Q45	“We employed three annotators who are experts in linguistics.” (p.4)
Q46	“To ensure the understanding of the tasks and annotation scheme, we took a sample of the dataset and asked each annotator to annotate them as per the prepared set of guidelines.” (p.4)
Q47	“Following this, every annotation was discussed in the presence of the annotators and an expert therapist as moderator to ensure consistency.” (p.4)
Q49	“We obtain the inter-rater agreement score of 0.7234 – which falls under the 'substantial' [7] category.” (p.4)
Q50	“Annotators are in the age group of 25-35, with 2-10 years of professional experience.” (p.4)
Q95	“Moreover, we consulted with mental-health professionals and linguists in preparing the annotation guideline.” (p.8)
Q96	“However, the annotator's bias cannot be ruled out completely.” (p.8)

Strengths

- Multi-annotator process with calibration exercise and expert-therapist moderation [Q46, Q47]
- Substantial aggregate inter-annotator agreement (Kappa = 0.7234) [Q49]
- Mental health professional and linguist consultation during guideline preparation [Q95]
- Authors transparently acknowledge residual annotator bias risk [Q96]

ID	Evidence
Q46	“To ensure the understanding of the tasks and annotation scheme, we took a sample of the dataset and asked each annotator to annotate them as per the prepared set of guidelines.” (p.4)
Q47	“Following this, every annotation was discussed in the presence of the annotators and an expert therapist as moderator to ensure consistency.” (p.4)
Q49	“We obtain the inter-rater agreement score of 0.7234 – which falls under the 'substantial' [7] category.” (p.4)
Q95	“Moreover, we consulted with mental-health professionals and linguists in preparing the annotation guideline.” (p.8)
Q96	“However, the annotator's bias cannot be ruled out completely.” (p.8)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Partially — labels reflect US clinical communication norms as encoded by linguistically-trained annotators with therapist moderation; South Asian and peer/crisis stakeholder perspectives are not documented as represented.

OC-2: Risk is bounded for in-scope clinical structured therapy (user judges A4 risk low) but unverified for South Asian clinical contexts, peer support practitioners, and crisis workers, none of whom are documented as annotators.

OC-3: Limited annotator documentation [Q50] — three linguists, ages 25–35, 2–10 years experience, expert-therapist moderator. No geographic, cultural, gender, or clinical-affiliation breakdown.

OC-4: Re-annotation by representative regional pool (South Asian clinicians, peer support workers, crisis staff) is not documented. The user's elicitation reasoning (A4) explicitly accepts this on grounds that labels are communicative, not clinical.

OC-5: Aggregation method is consensus discussion moderated by expert therapist [Q47]; this could erase minority annotator perspectives (e.g., on ambiguous ACK boundaries), though no minority-perspective tracking is reported.

OC-6: Documented issues: aggregate-only Kappa with no per-label or per-session-type breakdown; no cross-cultural annotator validation; complementary label pairs explicitly known to be challenging [Q51, Q52]; explicit acknowledgment of unresolved annotator bias [Q96].

ID	Evidence
Q45	“We employed three annotators who are experts in linguistics.” (p.4)
Q47	“Following this, every annotation was discussed in the presence of the annotators and an expert therapist as moderator to ensure consistency.” (p.4)
Q49	“We obtain the inter-rater agreement score of 0.7234 – which falls under the 'substantial' [7] category.” (p.4)
Q50	“Annotators are in the age group of 25-35, with 2-10 years of professional experience.” (p.4)
Q51	“The broad definitions of speaker-initiative and speaker-responsive dialogue-act label pairs, IRQ & ID; ORQ & OD, and CRQ & CD, seem complementary in nature.” (p.4)
Q52	“However, the dataset does not support the above view entirely.” (p.4)
Q95	“Moreover, we consulted with mental-health professionals and linguists in preparing the annotation guideline.” (p.8)
Q96	“However, the annotator's bias cannot be ruled out completely.” (p.8)

Information Gaps

- Per-label and per-session-type Kappa breakdown [deferred — not in available paper sections]
- Annotator geographic, cultural, and clinical-background details
- Whether South Asian clinicians or peer/crisis workers were consulted at any stage

Requires Expert Verification

- Whether South Asian psycholinguists would label ambiguous ACK/GC utterances consistently with HOPE annotators
- Whether peer support practitioners would assign the same labels to peer-register utterances

Remediation

Gap 1: No documented South Asian, peer-support, or crisis-worker annotators; aggregate-only Kappa of 0.7234 [Q49] cannot reveal cross-cultural or cross-session-type calibration.

Recommendation: Convene a small expert panel of South Asian clinical psychologists, peer-support practitioners, and crisis-intervention staff to re-annotate a stratified evaluation subset. Compute disagreement rates relative to HOPE's labels and use these to bound deployment confidence by stakeholder type.

ID	Evidence
Q49	“We obtain the inter-rater agreement score of 0.7234 – which falls under the 'substantial' [7] category.” (p.4)

Output Form (OF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: *HOPE* | ***Context:*** *US and South Asian Mental Health Counseling Practitioners (OMHC/Teletherapy)*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output form is single-label classification per utterance [Q14, Q30] with macro-F1, weighted-F1, and accuracy metrics [Q63] plus label-wise F1 reporting [Q76]. Statistical significance via T-test at >95% confidence is reported [Q88], and 3-fold cross-validation results are consistent with split-based results [Q87]. The output modality (categorical labels over text utterances) directly matches the deployment's expected output for session quality assessment and counselor feedback. Macro-F1 weights all classes equally regardless of frequency, appropriate for the imbalanced label distribution. The user marks OF as LOWER priority (form is well matched). Minor concerns: macro-F1 does not surface per-label safety-critical performance directly (e.g., crisis-relevant IRQ/YNQ behavior is averaged in); error analysis [Q89, Q90] is on in-distribution data only and does not assess behavior on out-of-scope (peer/crisis) utterances. Score of 4 reflects strong form alignment with minor diagnostic limitations.

ID	Evidence
Q14	“Each utterance in the dataset is tagged with one of the 12 counseling-aligned dialogue-act labels (c.f. Section 3).” (p.2)
Q30	“Sometimes, an utterance can have multiple dialog-acts; however, they are rare in the annotated dataset. Hence, for simplicity, we consider only one (primary) label for each utterance.” (p.3)
Q63	“To measure the performances of SPARTA and other baseline systems, we compute macro-F1, weighted-F1, and accuracy scores.” (p.6)
Q76	“We also present the label-wise performance of SPARTA in Table 4.” (p.7)
Q87	“Moreover, we also report the mean of the 3-fold cross-validation results for both SPARTA-MHA and SPARTA-TAA, and the results are consistent with the train-val-test split case.” (p.7)
Q88	“We also perform statistical significance T-test comparing SPARTA-TAA and the best performing baseline (CASA). We observe that our results are significant with > 95% confidence across macro-F1 (p-value= 0.009), weighted-F1 (p-value= 0.014), and accuracy (p-value= 0.048) values.” (p.7)
Q89	“Error Analysis: In this section, we present two-way error analyses of SPARTA in terms of quantitative and qualitative evaluations.” (p.7)
Q90	“Quantitative analysis: We report the confusion matrix for SPARTA-TAA in Figure 4.” (p.7)

Strengths

- Output modality (categorical labels) directly matches deployment use case for flow classification [Q14, Q63]
- Multiple complementary metrics reported (macro-F1, weighted-F1, accuracy) [Q63] including per-label F1 [Q76]
- Statistical significance testing [Q88] and 3-fold cross-validation [Q87] support reliability claims
- Quantitative and qualitative error analysis [Q89, Q90, Q91] provides interpretable failure-mode information

ID	Evidence
Q14	“Each utterance in the dataset is tagged with one of the 12 counseling-aligned dialogue-act labels (c.f. Section 3).” (p.2)
Q63	“To measure the performances of SPARTA and other baseline systems, we compute macro-F1, weighted-F1, and accuracy scores.” (p.6)
Q76	“We also present the label-wise performance of SPARTA in Table 4.” (p.7)

Q87	"Moreover, we also report the mean of the 3-fold cross-validation results for both SPARTA-MHA and SPARTA-TAA, and the results are consistent with the train-val-test split case." (p.7)
Q88	"We also perform statistical significance T-test comparing SPARTA-TAA and the best performing baseline (CASA). We observe that our results are significant with > 95% confidence across macro-F1 (p-value= 0.009), weighted-F1 (p-value= 0.014), and accuracy (p-value= 0.048) values." (p.7)
Q89	"Error Analysis: In this section, we present two-way error analyses of SPARTA in terms of quantitative and qualitative evaluations." (p.7)
Q90	"Quantitative analysis: We report the confusion matrix for SPARTA-TAA in Figure 4." (p.7)
Q91	"We observe three pairs with significant error-rates (≥ 25%) – YNQ:IRQ (26%), OD:ID (43%), ID:ACK (28%)." (p.7)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Yes — categorical-label output over text utterances matches deployment requirements (session flow classification for feedback/supervision).

OF-2: Not applicable — deployment is text-only; no TTS requirement indicated in deployment context.

OF-3: Not applicable as a primary concern — practitioners (not therapy clients) are the system's users; literacy is high in the practitioner population.

OF-4: Minor gaps: no separate evaluation of safety-critical label categories; no out-of-distribution evaluation framework for peer/crisis utterances; macro-F1 averaging may mask label-specific failure modes that matter for clinical review.

ID	Evidence
Q14	"Each utterance in the dataset is tagged with one of the 12 counseling-aligned dialogue-act labels (c.f. Section 3)." (p.2)
Q30	"Sometimes, an utterance can have multiple dialog-acts; however, they are rare in the annotated dataset. Hence, for simplicity, we consider only one (primary) label for each utterance." (p.3)
Q63	"To measure the performances of SPARTA and other baseline systems, we compute macro-F1, weighted-F1, and accuracy scores." (p.6)
Q76	"We also present the label-wise performance of SPARTA in Table 4." (p.7)

Information Gaps

- No reported evaluation framework for safety-critical label subsets
- No reported out-of-distribution behavior on peer/crisis-register utterances

Requires Expert Verification

- Whether macro-F1 at the reported 60.29 threshold is sufficient for supervisory feedback in clinical training contexts

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Remediation

Gap 1: Macro-F1 averaging masks safety-critical label performance; no out-of-distribution evaluation for peer/crisis utterances [Q63, Q76].

Recommendation: Report and monitor per-label F1 (especially IRQ, YNQ, ACK, GC) separately in deployment dashboards. Add an out-of-distribution detection layer that flags utterances likely outside the trained label distribution rather than forcing a label assignment, particularly for crisis-context deployment.

ID	Evidence
Q63	“To measure the performances of SPARTA and other baseline systems, we compute macro-F1, weighted-F1, and accuracy scores.” (p.6)
Q76	“We also present the label-wise performance of SPARTA in Table 4.” (p.7)